

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
OPEN TEST – I
PAPER –2
TEST DATE: 23-01-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	-1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. 10^{25} atoms of helium in a container of volume 27 liter are in random motion. Assume speed of atoms is same and equal to 2×10^3 m/s. Mass of He atom is 6.65×10^{-27} kg and each collision is elastic.

- (A) Pressure of the gas is 3.3×10^6 N/m²
 (B) Pressure of the gas is 6.6×10^6 N/m²
 (C) Number of atoms hitting wall of 1 m² area in a second is 1.23×10^{29}
 (D) Number of atoms hitting wall of 1 m² area in a second is 6.17×10^{28}

Ans. A, C

Sol. no. of atoms per m³ = $\frac{10^{25}}{27 \times 10^{-3}} = \frac{10^{28}}{27}$

Assuming a cube of 1 m³ volume frequency of collision is n

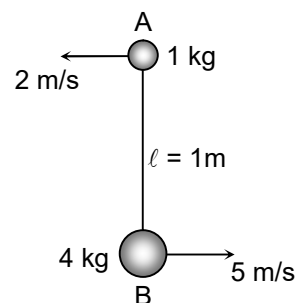
∴ No. of atoms colliding 1 m² in one second

$$n = \left(\frac{10^{28}}{27} \right) \frac{1}{6} \times 2000 = 1.23 \times 10^{29}$$

Pressure P = 2m_{He}n

$$= 2 \times 6.65 \times 10^{-27} \times 2000 \times 1.23 \times 10^{29} = 3.3 \times 10^6 \text{ N/m}^2$$

2. Two balls A and B of mass 1 kg and 4 kg respectively are connected by an ideal string and placed on a smooth horizontal surface. A and B are given velocity perpendicular to the string connecting them as shown in the figure. Motion of balls are in horizontal plane.



- (A) Acceleration of ball A is 39.2 m/s²
 (B) Acceleration of ball B is 9.8 m/s²
 (C) Tension in the string is 39.2 N
 (D) Tension in the string is 4 N

Ans. A, B, C

Sol. Let acceleration of B is a towards A
 Motion of A with respect to B

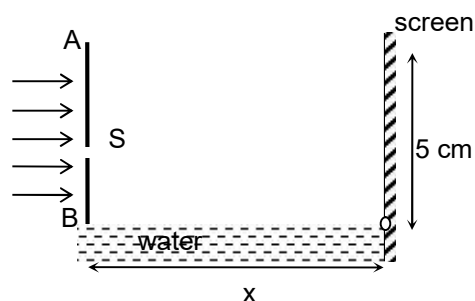
$$\begin{aligned}
 T + 1 \cdot a &= 1 \times \frac{7^2}{1} \\
 \vec{a}_{A,g} &= \vec{a}_{A,B} + \vec{a}_{B,g} \\
 &= \frac{7^2}{1} \downarrow + a \uparrow \\
 &= (49 - a) \downarrow
 \end{aligned}$$

Motion of B with respect to A

$$T + 4(49 - a) = 4 \times \frac{7^2}{1}$$

Solving we get $T = 39.2 \text{ N}$ and $a = 9.8 \text{ m/s}^2$

3. Interference pattern experiment is done on the surface of water by using a slit S in an opaque plate AB and screen is placed at 50 cm from slit. Slit is 1 cm above water surface and light of wavelength $\lambda = 6000 \text{ \AA}$ is falling over it. The plate is oscillating according to equation $X = A \sin \omega t$ along the line BO, where $A = 1 \text{ cm}$, $\omega = 100 \text{ rad/s}$. Point O is just above the water surface and another point P is 5 cm above the water surface. Plate AB is just above the water surface.



- (A) Intensity at point O is minimum
 (B) Intensity at point O is maximum
 (C) Maximum speed of fringe at point P is 10 cm/s
 (D) Maximum speed of fringe at point P is 5 cm/s

Ans. A, C

Sol. Reflection will take place from water surface which will introduce $\lambda/2$ path difference. At distance y above point O, path difference

$$\Delta x = \frac{dy}{D} + \frac{\lambda}{2}$$

for $y = 0$, $\Delta x = \frac{\lambda}{2} \Rightarrow$ at point O, I is minimum.

$$\Delta x = n\lambda = \frac{dy}{D} + \frac{\lambda}{2}$$

$$\Rightarrow y = \frac{D\lambda}{d} \left(n - \frac{1}{2} \right)$$

$$\frac{dy}{dt} = \frac{\lambda}{d} \left(n - \frac{1}{2} \right) \frac{dD}{dt} = \frac{y}{D} A\omega \cos \omega t$$

$$\left(\frac{dy}{dt} \right)_{\max} = \frac{A\omega y}{D} = \frac{1 \text{ cm} \times 100 \times 5 \text{ cm}}{50 \text{ cm}} = 10 \text{ cm/s}$$

4. Two particles of equal mass m interact with each other by a force for which potential energy function $U = U_0 \left(\frac{1}{r^{12}} - \frac{2}{r^6} \right)$

Where U_0 is a constant, r is separation between the particles. Equilibrium separation between them is r_0 . When their separation slightly changes from equilibrium separation they start oscillating with angular frequency ω . Then choose the correct option(s).

- (A) $r_0 = 1$
 (B) $r_0 = (2)^{-1/6}$
 (C) $\omega = \sqrt{\frac{144U_0}{m}}$
 (D) $\omega = \sqrt{\frac{72U_0}{m}}$

Ans. A, C

Sol. $F = -\frac{dU}{dr} = -U_0 \left(-\frac{12}{r^{13}} + \frac{12}{r^7} \right)$

$F = 0 \Rightarrow r_0 = 1$

If slightly displaced by Δr then

$r = r_0 + \Delta r$

$= 1 + \Delta r$

$F = -U_0 \left[\frac{-12}{(1 + \Delta r)^{13}} + \frac{12}{(1 + \Delta r)^7} \right] = -72U_0 \Delta r$

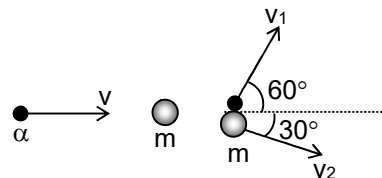
$a = \frac{F}{(m/2)} \Rightarrow a = -\left(\frac{144U_0}{m} \right) \Delta r \Rightarrow \omega = \sqrt{\frac{144U_0}{m}}$

5. α -particle collides elastically with a nucleus of gas which is filled in a chamber. After collision α -particle deviates at 60° and nucleus at 30° with the direction of motion of incident α -particle. Then choose the correct option(s).

- (A) Mass number of gas filled in chamber is 4
 (B) Mass number of gas filled in chamber is 2
 (C) Ratio of the speed of struck nucleus to the speed of incident α -particle is $\frac{1}{\sqrt{3}}$
 (D) Ratio of the speed of struck nucleus to the speed of incident α -particle is $\frac{\sqrt{3}}{2}$

Ans. A, D

Sol. mass of α -particle is 4
 $\therefore$ From COM
 $4v = 4v_1 \cos 60^\circ + mv_2 \cos 30^\circ$
 And $0 = 4v_1 \sin 60^\circ - mv_2 \sin 30^\circ$
 Conservation of energy
 $\frac{1}{2} \times 4v^2 = \frac{1}{2} \times 4v_1^2 + \frac{1}{2} mv_2^2$
 Solving $m = 4$
 and $\frac{v_2}{v} = \frac{\sqrt{3}}{2}$



6. A particle of charge q and mass m is dropped from rest. It falls under the effect of uniform gravity where acceleration due to gravity is g . There exists a uniform horizontal magnetic field B . Choose the correct statement(s).

- (A) After falling through certain distance the particle starts climbing up
 (B) Speed of particle when it starts climbing is $\frac{2mg}{qB}$
 (C) Depth after which particle starts climbing is $\frac{2m^2g}{q^2B^2}$
 (D) Speed of particle when it starts climbing is $\frac{mg}{qB}$

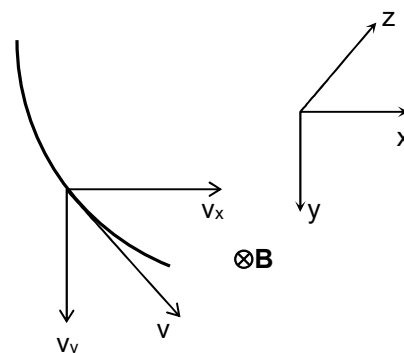
Ans. A, B, C

Sol. $m \left(\frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} \right) = mg \hat{j} + q(v_x \hat{i} + v_y \hat{j}) \times B \hat{k}$
 $\Rightarrow \frac{dv_x}{dt} = \frac{qB}{m} v_y$ and $\frac{dv_y}{dt} = g - \frac{qB}{m} v_x$
 Solving $v_x = \frac{mg}{qB} (1 - \cos \omega t)$ and $v_y = \frac{mg}{qB} \sin \omega t$
 Where $\omega = \frac{qB}{m}$
 $v_y = 0 \Rightarrow \omega t = \pi$
 $\Rightarrow v_x = \frac{2mg}{qB}$

Conservation of energy

$$mgh_{\max} = \frac{1}{2} m \left(\frac{2mg}{qB} \right)^2$$

$$\Rightarrow h_{\max} = \frac{2m^2g}{q^2B^2}$$

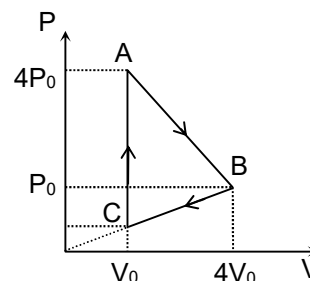


Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

One mole of an ideal monoatomic gas is taken through cyclic process ABCA as shown in the figure. Temperature in state A and B are equal to 400 K.



7. What is the maximum temperature achieved in cyclic process?

- (A) 400 K
- (B) 625 K
- (C) 450 K
- (D) 550 K

Ans. B

Sol. $T = \frac{PV}{nR}$

For $T_{\max}$ product of PV must be maximum T_c is minimum.

$$T_A = T_B = 400 \text{ K}$$

$$\Rightarrow \frac{4P_0 V_0}{R} = 400 \Rightarrow \frac{P_0 V_0}{R} = 100$$

A $\rightarrow$ B process

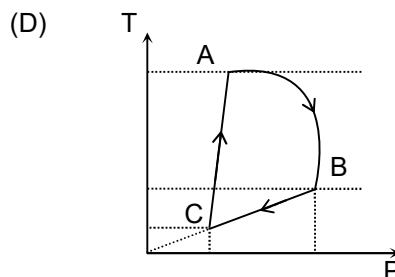
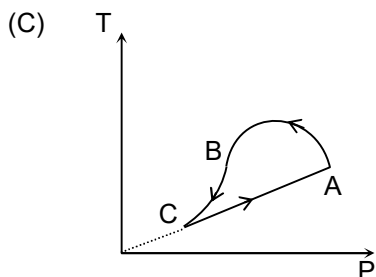
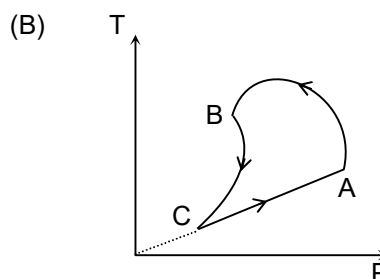
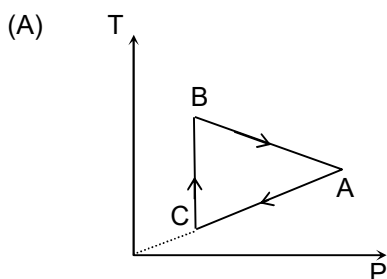
$$P = -\frac{P_0}{V_0} V + 5P_0$$

$$\frac{RT}{V} = P \Rightarrow T = \frac{-P_0}{RV_0} V^2 + \frac{5P_0}{R} V$$

$$\frac{dT}{dV} = 0 \Rightarrow V = \frac{5}{2} V_0 \text{ and } T_{\max} = -\frac{P_0}{RV_0} \left(\frac{5}{2} V_0 \right)^2 + \frac{5P_0}{R} \left(\frac{5}{2} V_0 \right)$$

$$T_{\max} = \frac{25 P_0 V_0}{4 R} = 625 \text{ K}$$

8. The cyclic process is best represented by T-P diagram



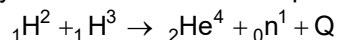
Ans. C

Sol. Use the relation from the given graph, and make the required graph

Paragraph for Question Nos. 09 and 10

Experiments are going on to produce large scale energy by using Deuterium (${}_1\text{H}^2$) and Tritium (${}_1\text{H}^3$).

They are using laser beams to compress the material. Fuel pellets are made by mixing equal number of atoms of ${}_1\text{H}^3$ and ${}_1\text{H}^2$. Such fuel pellets are spheres of radius $r = 200 \mu\text{m}$ and its density is 200 kg/m^3 . When laser beams hit the pellet, it gets compressed and its density is increased by 1000 times. Assume that only 10% of the atoms take part in fusion reaction



mass of ${}_1\text{H}^2 = 2.0141 \text{ u}$

mass of ${}_1\text{H}^3 = 3.0160 \text{ u}$

mass of ${}_2\text{He}^4 = 4.0026 \text{ u}$

mass of ${}_0\text{n}^1 = 1.0087 \text{ u}$

(Take, $1 \text{ amu} = 931 \text{ MeV}/c^2$)

9. The Q-value of this nuclear reaction is

(A) 3.27 MeV

(B) 4.03 MeV

(C) 956 MeV

(D) 17.5 MeV

Ans. D

10. The energy produced in laser action of a pallet is

- (A) 10^8 J
 (B) 10^5 J
 (C) 10^3 J
 (D) 10^6 J

Ans. B

Sol. (for Q. 9-10)

$$Q = (m_{1\text{H}^2} + m_{1\text{H}^3} - m_{2\text{He}^4} - m_{0\text{n}^1}) \times 931 \text{ MeV} = 17.5 \text{ MeV}$$

Let n is number of atoms per unit volume then

$$200 \times 10^3 = \frac{n}{2} m_d + \frac{n}{2} m_T = \frac{n}{2} \left(\frac{M_d}{N_A} + \frac{M_T}{N_A} \right)$$

$$n = \frac{2N_A(2 \times 10^5)}{M_d + M_T} = \frac{2 \times 6.02 \times 10^{23} \times 2 \times 10^5}{(2 \times 10^{-3} + 3 \times 10^{-3})} = 4.8 \times 10^{31} / \text{m}^3$$

$$\text{Volume of pallet} = \frac{4}{3} \pi (20 \times 10^{-6})^3$$

$$\text{Number of atoms in a pallet} = 4.8 \times 10^{31} \times \frac{4}{3} \times 3.14 \times (20 \times 10^{-6})^3 = 1.61 \times 10^{18}$$

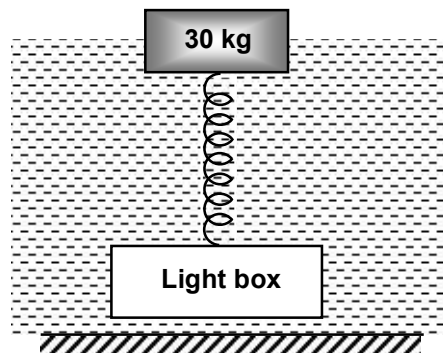
Energy produced in one laser action on a pallet is

$$\frac{1.61 \times 10^{17}}{2} \times 17.5 \times 10^6 \times 1.6 \times 10^{-19} = 2.2 \times 10^5 \text{ J}$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. A block of mass 30 kg and an empty and light Box of equal volume are attached by an ideal spring of force constant $k = 400 \text{ N/m}$. The whole system is floating in water of density (ρ_w) = 10^3 kg/m^3 . The upper block is half submerged into water. Take the value of acceleration due to gravity $g = 10 \text{ m/s}^2$. If the compression in spring is 10k cm. Then find the value of k .



Ans. 5

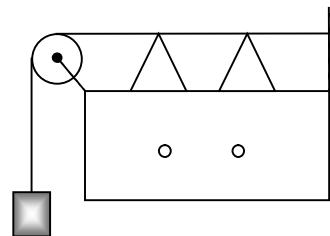
Sol. $30g = kx + \frac{V}{2} \rho_w g$

And

$$kx = V \rho_w g$$

Solving, $x = 50 \text{ cm}$

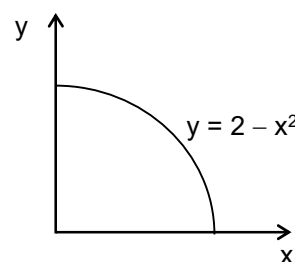
12. A tuning fork produces 3 beats with a sonometer wire of length 80 cm. When separation between the bridges is increased by 1 cm, tuning fork resonates with sonometer wire. If frequency of tuning fork is 80P Hz. Find the value of P.



Ans. 3

Sol.
$$\frac{v}{2 \times 80} - f_T = 3$$
$$f_T = \frac{v}{2 \times 81} = 3 \times 80 \text{ Hz}$$

13. A bead of charge q and mass m is free to move on a smooth fixed insulating wire, which is in the shape of parabola $y = (2 - x^2)$ located in a vertical plane. Uniform gravity $\vec{g} = 10(-\hat{j}) \text{ m/s}^2$ is present in the space. The bead is pushed slightly on the wire from $(0, 2)\text{m}$, it reaches on the ground ($y = 0$) with speed v . Electric field $\vec{E} = \alpha(x\hat{i} + y\hat{j}) \text{ N/C}$ is present in space and $\frac{q\alpha}{m} = 2$ (in SI unit). Find the speed v (in m/s).



Ans. 6

Sol. Work energy theorem

$$\int (m\vec{g} + q\vec{E}) \cdot (dx\hat{i} + dy\hat{j}) = \frac{1}{2}mv^2 - 0$$
$$\int [mg(-\hat{j}) + q\alpha(x\hat{i} + y\hat{j})] (dx\hat{i} + dy\hat{j}) = \frac{1}{2}mv^2$$
$$-mg \int_2^0 dy + q\alpha \int_0^{\sqrt{2}} x dx + q\alpha \int_2^0 y dy = \frac{1}{2}mv^2$$
$$\Rightarrow v = 6 \text{ m/s}$$

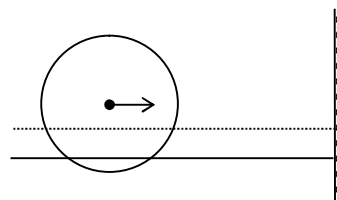
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

A sphere of radius $R = 5 \text{ cm}$ is rolling without slipping with a velocity 10 m/s on two parallel, horizontal fixed rods as shown in the figure. The separation between the rods is $\ell = 2\sqrt{24} \text{ cm}$. There is friction between the rods and sphere and coefficient of friction $\mu = 0.3$. The sphere collides with a smooth vertical wall elastically. It returns back and again collides with a velocity v_2 . Similarly it collides with the wall again with a velocity v_3 .



14. The value of v_2 (in m/s) is

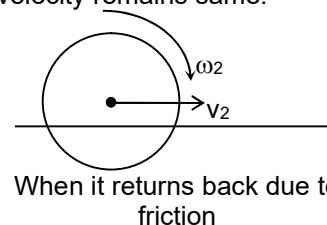
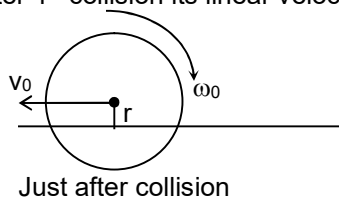
Ans. 8.18

15. The value of v_3 (in m/s) is

Ans. 6.69

Sol. (for Q. 14-15)

After 1st collision its linear velocity becomes opposite but angular velocity remains same.



Condition of rolling on rods $v_0 = r\omega_0$

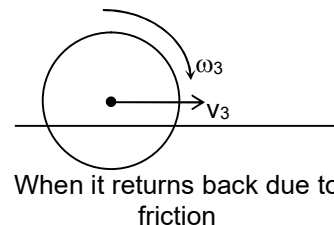
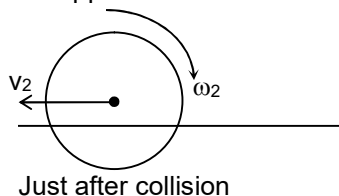
$$r = 1 \Rightarrow v_0 = \omega_0$$

and $v_2 = \omega_2$

Applying COAM about mid point in the plane of rods

$$-mv_0r + \frac{2}{5}mR^2\omega_0 = mv_2r + \frac{2}{5}mR^2\omega_2$$

$$\Rightarrow v_2 = \frac{9}{11}v_0$$



Conservation of angular momentum provides

$$v_3 = \frac{9}{11}v_2 = \frac{9}{11} \times \frac{9}{11}v_0$$

By putting the values

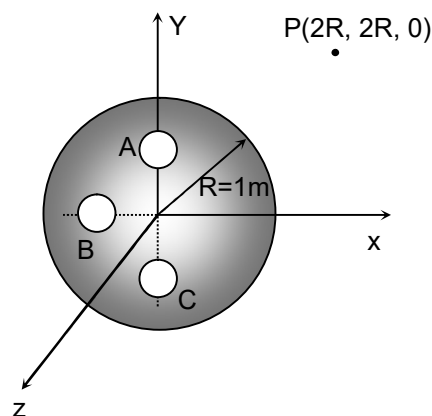
$$v_2 = 8.18 \text{ m/s}$$

$$v_3 = 6.69 \text{ m/s}$$

Question Stem for Question Nos. 16 and 17

Question Stem

A metal sphere with its centre at origin has radius $R = 1\text{m}$. It has three spherical cavities each of radius $R/4$ and their centres are at $\left(0, \frac{R}{2}, 0\right)$, $\left(-\frac{R}{2}, 0, 0\right)$ and $\left(0, -\frac{R}{2}, 0\right)$ respectively. Initially cavity A and B have some non zero charges and cavity C is empty. The electric field at point P is zero.



16. When charge from cavity A is removed, electric field at point $P(2R, 2R, 0)$ is found to be $1 \frac{\text{kV}}{\text{m}}$. The magnitude of charge present in the cavity A (in μC) was

Ans. 0.89

17. By keeping the charges same in the cavities A and B and the charge of $2 \mu\text{C}$ is placed at the centre of cavity C. The electric field at point P (in $\frac{\text{kV}}{\text{m}}$) is

Ans. 2.25

Sol. (for Q. 16-17)

$E = 0 \Rightarrow$ net charge on sphere is zero. If one is removed, field $E = \frac{kQ}{r^2}$

$$1 \times 10^3 = \frac{9 \times 10^9 \times q \times 10^{-6}}{8R^2}$$

$$R = 1 \Rightarrow q = 0.89 \mu\text{C}$$

Now field is due to charge induced on surface due to charge present in cavity C

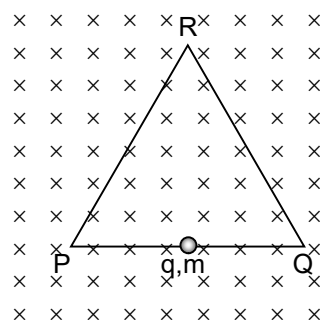
$$E = \frac{k|q_C|}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6}}{8R^2}$$

$$R = 1 \Rightarrow E = 2.25 \frac{\text{kV}}{\text{m}}$$

Question Stem for Question Nos. 18 and 19

Question Stem

A uniform time varying magnetic field $B = B_0 t$ is present in a cylindrical region. A fixed wire frame PQR of equilateral triangular shape has slight curve at its corners. Side length of frame is $\ell = 2\sqrt{3}$ m and it is made of insulating material. A charged bead of charge q and mass m can move on the wire frame without friction is released from rest at mid point of PQ. It is assumed that bead moves through corners of frame without any loss of kinetic energy. The bead moves on the frame and again comes at initial position in time t_0 and with a speed v_0 . It is given that $\frac{qB_0}{m} = \sqrt{3}$ (in SI unit).



18. The value of time t_0 (in sec) is

Ans. 4.90

19. The value of speed v_0 (in m/s) is

Ans. 4.24

Sol. (for Q. 18-19)

$$E 2\pi r = \pi r^2 \frac{dB}{dt}$$

$$\Rightarrow E = \frac{B_0 r}{2}$$

$$a_t = \frac{qE}{m} \cos \theta = \frac{qB_0 D}{2m}$$

$$3\ell = \frac{1}{2} \frac{qB_0 D}{2m} t^2$$

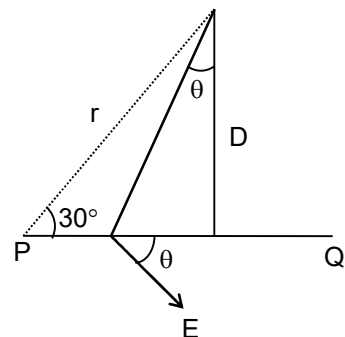
$$t = \sqrt{\frac{12\ell}{\left(\frac{qB_0}{m}\right) D}}$$

...(i)

$$\frac{D}{(\ell/2)} = \tan 30^\circ \Rightarrow D = \frac{\ell}{2\sqrt{3}} = 1$$

$$t = \sqrt{24} = 4.90 \text{ s}$$

$$v^2 = 2a_t(3\ell) = 2 \times \frac{qB_0 D}{2m} \times 3 \times 2\sqrt{3} = 4.24 \text{ m/s}$$



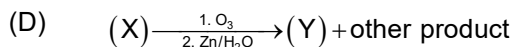
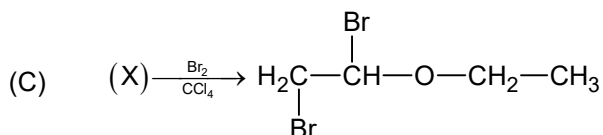
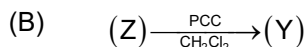
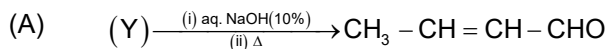
Chemistry

PART – II

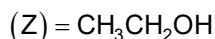
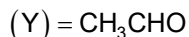
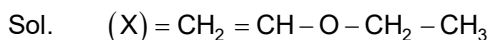
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

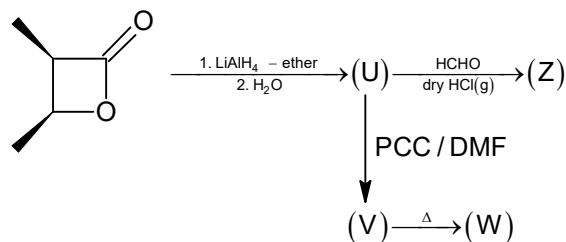
20. Compound (X), C_4H_8O decolorizes Baeyer's reagent. It undergoes hydrolysis on reaction with dil. H_2SO_4 and produces (Y) and (Z). Both (Y) and (Z) give iodoform test positive. Only (Y) gives Tollen's silver mirror test positive



Ans. A, B, C



21.



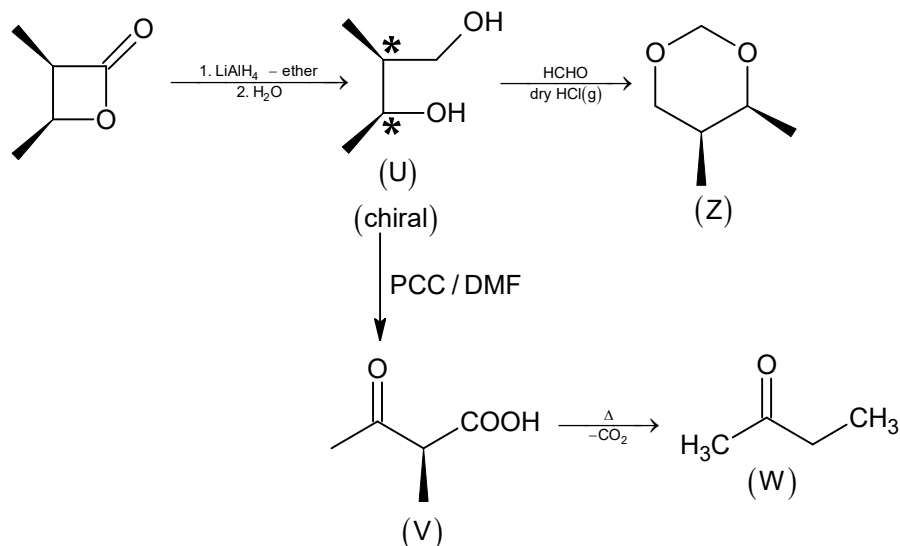
(Where DMF = N,N - dimethyl formamide)

Identify the correct statement (s) among the following:

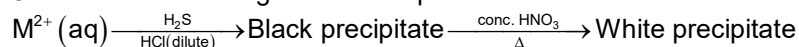
- (A) 'U' is optically active.
 (B) 'V' gives effervescence with $NaHCO_3$.
 (C) Molecular formula of 'Z' $C_6H_{12}O_2$.
 (D) 'W' gives iodoform test.

Ans. A, B, C, D

Sol.



22. Consider the following reaction sequence:


 Which of the following metal ion(s) can be M^{2+} ?

- (A) Hg^{2+}
 (B) Ni^{2+}
 (C) Cu^{2+}
 (D) Pb^{2+}

Ans. C, D

 Sol. Only IInd group cations form ppt. of sulphide with H_2S in presence of dil. HCl.

So, Ni^{2+} cannot be M^{2+} as it belongs to IVth group. Also, HgS does not dissolve in nitric acid. So, Hg^{2+} is also not possible. CuS and PbS are black precipitates which dissolve in nitric acid form colloidal sulphur, a white ppt.

23. An ideal gas is subjected to three different type of adiabatic processes from same initial state to same final volume as given below:

Expansion	Work	Final temperature	Internal energy change	Enthalpy change
Reversible	W	T_2	ΔU	ΔH
Irreversible	W'	T'_2	$\Delta U'$	$\Delta H'$
Free	W''	T''_2	$\Delta U''$	$\Delta H''$

Which of the following relation(s) is/are correct?

- (A) $T_2 < T'_2 < T''_2$
 (B) $W < W' < W''$ and $|W| > |W'| > |W''|$

(C) $\Delta U < \Delta U' < \Delta U''$

(D) $\Delta H < \Delta H' < \Delta H''$

Ans. A, B, C, D

Sol. (i) Free expansion of an ideal gas is isothermal as well as adiabatic irreversible process. So, $q = 0, W'' = 0, \Delta U'' = 0, \Delta H'' = 0$.

(ii) For the same change of volume, magnitude of work done in reversible expansion is more than irreversible expansion.

So, $|W| > |W'|$ or $W < W'$ (with sign)

$\Delta U = W$ (for adiabatic)

$\Delta U < \Delta U'$

Also, $\Delta H < \Delta H'$

$P_{\text{final}} = \frac{nRT_f}{V_f}$

So, $T_2 < T_2' < T_2''$

24. The nitrogen containing product obtained by heating lead nitrate:

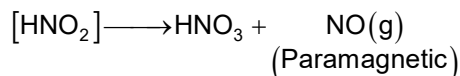
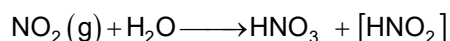
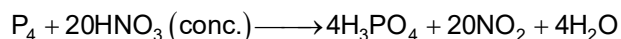
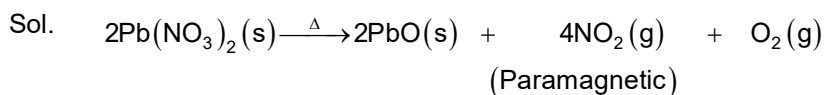
(A) can also be obtained by reacting white phosphorus with conc. HNO_3 .

(B) is diamagnetic.

(C) contains sp^2 hybridized nitrogen atom.

(D) liberates a paramagnetic gas when reacted with water.

Ans. A, C, D



25. $\text{B}_3\text{N}_3\text{H}_6$ + solution of hydrochloric acid $\longrightarrow$?

Select 'INCORRECT' statement(s) about the above reaction:

(A) No reaction

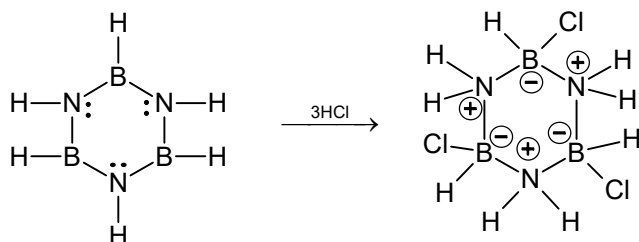
(B) $\text{B}_3\text{N}_3\text{H}_6$ shows substitution reaction and forms $\text{B}_3\text{N}_3\text{Cl}_6$

(C) $\text{B}_3\text{N}_3\text{H}_6$ show addition reaction and produces $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$ in which 'Cl'-atoms are bonded to boron atoms.

(D) $\text{B}_3\text{N}_3\text{H}_6$ shows addition reaction and produces $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$ in which 'Cl'-atoms are bonded to nitrogen atoms.

Ans. A, B, D

Sol.

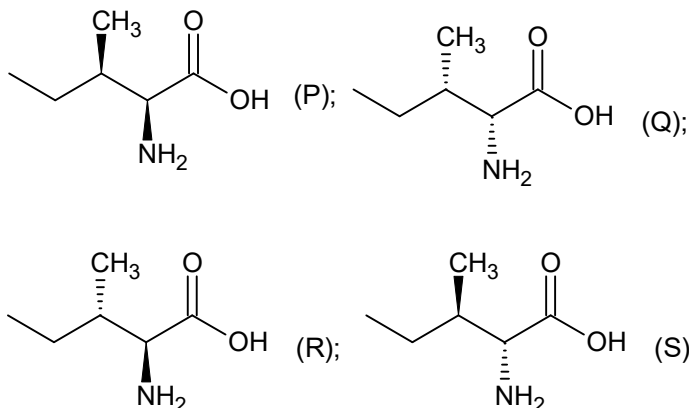


Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27

Consider the stereoisomer of iso-leucine labelled as below:



Consider the following data to answer the questions posted:

- Set molar Gibb's free energy of 'P', $G_m^\circ = 0 \text{ kJ mol}^{-1}$.
- The equilibrium constant (K_{eq}) for epimerization of (P) has the value of 1.38 (at 374 K).
- Epimerization of (R) into (Q) follows first order kinetics with rate constant 'k' (at 374 K) = $9.02 \times 10^{-5} \text{ hr}^{-1}$ and considered to be irreversible.

$$(iv) \text{ Diastereomeric excess is defined as } D_e = \frac{R - Q}{R + Q} \times 100.$$

Where, [R] and [Q] represents the concentrations of R and Q forms in their mixture.

26. What is the G_m° of 'Q'? [Given $\ln 1.38 = 0.322$]

- 0 kJ
- $-1.00 \times 10^3 \text{ kJ mol}^{-1}$
- $+1.00 \times 10^3 \text{ kJ mol}^{-1}$
- $-1.38 \text{ kJ mol}^{-1}$

Ans. A

Sol. Enantiomers have same molar free energy.

27. If 'R' is boiled for 1943 hours at 374 K, then it undergoes epimerization to form (Q). What is the value of Diastereomeric excess (D_e) after 1943 hours? ($e^{0.175} = 1.19$)

(A) 83.5%

(B) 12.5%

(C) 68.2%

(D) 19.6%

Ans. C

Sol.

	$R \xrightarrow{\text{order}=1} Q$	
at $t = 0$	1	0
after time 't'	$(1 - \alpha)$	α

$$9.02 \times 10^{-5} = \frac{1}{1943} \times \ln \frac{1}{(1 - \alpha)}$$

$$\ln \frac{1}{(1 - \alpha)} = 1943 \times 9.02 \times 10^{-5} = 0.175$$

$$\frac{1}{(1 - \alpha)} = e^{0.175} = 1.19$$

$$\therefore \alpha = 0.159$$

$$\text{So, } (R) = 1 - 0.159 = 0.841, (Q) = 0.159$$

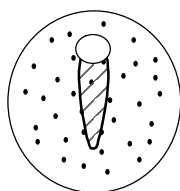
$$\text{So, Diastereomeric excess } (D_e) = \frac{[R] - [Q]}{[R] + [Q]} \times 100$$

$$D_e = \frac{(0.841 - 0.159)}{(0.841 + 0.159)} \times 100$$

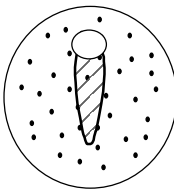
$$D_e = \frac{0.682}{1} \times 100 = 68.2\%$$

Paragraph for Question Nos. 28 and 29

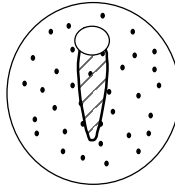
Rust indicator containing potassium hexacyano ferrate (II) solution was poured into the following glass dishes to cover the iron nails which were wrapped with different metal strips. The dishes were allowed to stand in air for some time.



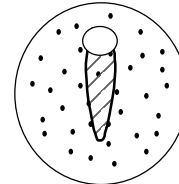
Silver strip
(Dish - 1)



Zinc strip
(Dish - 2)



Copper strip
(Dish - 3)



Magnesium strip
(Dish - 4)

28. If the iron nail rusts, what would the colour of rust indicator be around the nail?

- (A) Yellow
- (B) Brown
- (C) Red
- (D) Blue

Ans. D

Sol. In the rust, iron is present in +3 oxidation state and form blue solution with $K_4[Fe(CN)_6]$.

29. In which of the dishes would the iron nail rust?

- (A) Dish – 1 only
- (B) Dish – 2 only
- (C) Dishes – 1 and 3 only
- (D) Dishes – 2 and 4 only

Ans. C

Sol. Since Zn and Mg are more electropositive than iron. So, iron will not rust. But Cu and Ag are less electropositive than iron, so, can not reduce the rust.

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

30. A sample of camphor used in a method for determining of molar mass had melting point of 450 K. The melting point of a solution containing 0.5 gm camphor and 0.04 gm of an unknown non-volatile non-electrolyte compound was 430 K. The molar mass of camphor is 152 and its $\Delta H_{\text{fusion}} = 1.52 \text{ Kcal mol}^{-1}$. The molar mass of unknown compound in gm/mole is.....
 ($R = 2 \text{ cal mol}^{-1} \text{ K}^{-1}$).

Ans. 162

Sol. $K_f(\text{of camphor}) = \frac{R(T_f^\circ)^2 \times M_2}{1000 \times \Delta H_f^\circ} = \frac{2 \times 152 \times (450)^2}{1000 \times 1.52 \times 1000} = 40.5$

$$\Delta T_f = i \times K_f \times M$$

$$450 - 430 = 1 \times 40.5 \times \frac{0.04 \times 1000}{M_{\text{solute}} \times 0.5}$$

$$M_{\text{solute}} = \frac{0.04 \times 1000 \times 40.5}{0.5 \times 20} = 162$$

31. The sum of total numbers of isomers possible for the complexes $[\text{Co}(\text{ox})(\text{PMe}_3)_2(\text{NH}_3)(\text{Cl})]$ and $\text{K}_4\text{Fe}(\text{CN})_6$ is.....

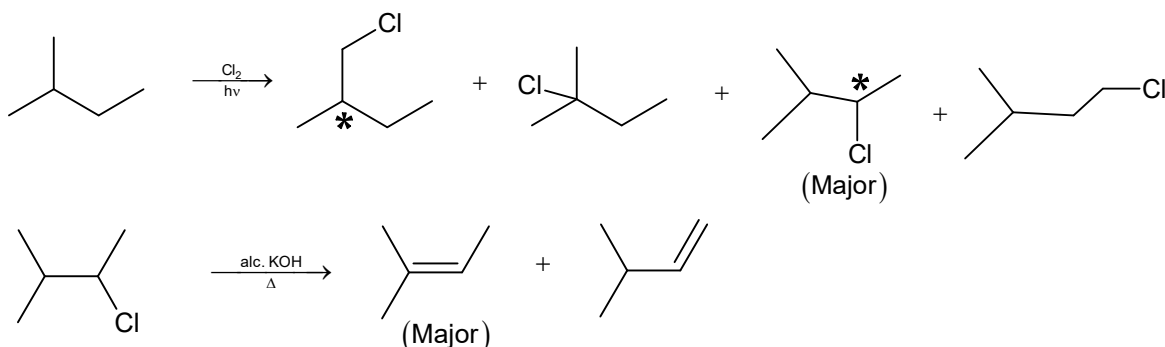
Ans. 16

Sol. Total isomers of $[\text{Co}(\text{ox})(\text{PMe}_3)_2(\text{NH}_3)(\text{Cl})]$ are 6 and total isomers of $\text{K}_4\text{Fe}(\text{CN})_6$ are 10.

32. 2-methyl butane on photochemical monochlorination gives 'x' number of isomeric products of the formula $\text{C}_4\text{H}_9\text{Cl}$. The number of product(s) obtained when major structural isomer of $\text{C}_4\text{H}_9\text{Cl}$ produced in the reaction is heated with KOH in presence of $\text{C}_2\text{H}_5\text{OH}$ is y. What is the value of $(x + y)$?

Ans. 8

Sol.



Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

An acid 'A' which is an important constituent of vinegar; on reaction with red phosphorous and Br_2 , gives a monobromo derivative (B) which on reaction with NH_3 gives white solid (C). However, ethanal on reaction with a mixture of ammonium chloride and sodium cyanide undergoes strecker synthesis to give a product which on acidic hydrolysis gives another high melting solid (D). The sum of total optical isomers of (C) and (D) are 'x' while 'y' is the total number of dipeptides (excluding stereoisomers) that a mixture of (C) and (D) can form.

33. The value of x is _____

Ans. 2.00

Sol. $\text{A} = \text{CH}_3\text{COOH}$; $\text{B} = \text{H}_2\text{C}(\text{Br})\text{COOH}$; $\text{C} = \text{H}_2\text{C}(\text{NH}_2)\text{COOH}$; $\text{D} = \text{H}_3\text{C}-\text{CH}(\text{NH}_2)\text{COOH}$

34. The value of y is _____

Ans. 4.00

Sol. Possible dipeptides are:

C – C ; D – D

C – D ; D – C

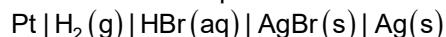
Question Stem for Question Nos. 35 and 36

Question Stem

Scheme 1: A person takes 6.1 gm of an antacid tablet comprising bicarbonate ion at 20.8% by mass. The volume of $\text{CO}_2(\text{g})$ evolved at (1 atm, 25°C) in the stomach be 'x' litres.

$$[R = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}]$$

Scheme 2: The standard cell potential of the following cell:



was measured over a range of temperatures, and the data was fitted as:

$$E_{\text{cell}}^{\circ} (\text{volts}) = 0.01 - 1 \times 10^{-4} (T - 298) - 2 \times 10^{-6} (T - 298)^2$$

Where 'T' is the absolute temperature. The standard enthalpy change of the cell reaction at 298 K is – 'y' kJ mol^{-1} corresponding to 1 Faraday charge [1 F = 96500 C].

35. The value of 'x' is _____

Ans. Range (0.50 – 0.52)

Sol. $\text{HCO}_3^- + \text{H}^+ \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$

$$\text{Moles of } \text{HCO}_3^- = \text{moles of } \text{CO}_2 = \frac{6.1}{61} \times \frac{20.8}{100} = 0.0208$$

$$PV = nRT$$

$$V = \frac{0.0208 \times 0.0821 \times 298}{1} = 0.5088 \text{ L} \approx 0.51 \text{ L}$$

36. Value of 'y' is _____

Ans. Range (3.80 – 3.85)

Sol. $(\Delta S_{298}^{\circ}) = +nF \left(\frac{\delta E^{\circ}}{\delta T} \right)_{P, 298}$

$$= 1 \times 96500 [-10^{-4}]$$

$$= -9.65 \text{ J / K}$$

$$\Delta G_{298}^{\circ} = -nFE_{\text{cell}}^{\circ}$$

$$\Delta G_{298}^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$$

$$\Delta H^{\circ} = \Delta G^{\circ} + T\Delta S^{\circ}$$

$$= -0.965 + 298(-9.65 \times 10^{-3})$$

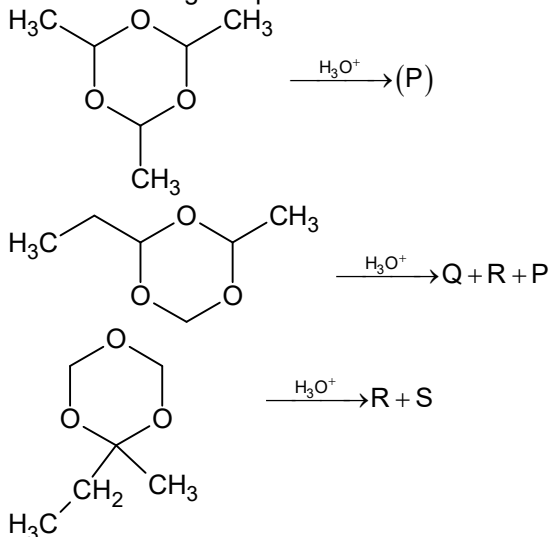
$$= -0.965 - 2.8757$$

$$\Delta H_{298}^{\circ} = -3.8407 \text{ kJ mol}^{-1}$$

Question Stem for Question Nos. 37 and 38

Question Stem

Three isomers of molecular formula ($C_6H_{12}O_3$) is shown below. On complete acidic hydrolysis P, Q, R and S are formed as organic products:



Let, the ratio of the number of products (P, Q, R, S) which gives haloform reaction to the co-ordination number of an atom in the top most layer of cubic close packed (ccp) structure be 'x' and let the ratio of packing fraction of diamond to the number of organic products (P, Q, R, S) which gives Tollen's test positive be 'y', ($\pi = 3.14, \sqrt{2} = 1.414, \sqrt{3} = 1.732$)

37. Value of 'x' is _____

Ans. 0.25

Sol. $P = CH_3 - CH = O$; $Q = CH_3 - CH_2 - CHO$;

$R = HCHO$; $S = CH_3 - CH_2 - \overset{\overset{O}{||}}{C} - CH_3$

$$\text{So, } x = \frac{2}{8} = 0.25$$

38. Value of 'y' is _____

Ans. Range (0.10 – 0.12)

Sol. $y = \frac{0.34}{3} = 0.11$

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Curves A, B, C and D are defined in the plane as follows;

$$A = \left\{ (x, y); \quad x^2 - y^2 = \frac{x}{x^2 + y^2} \right\}; B = \left\{ (x, y); \quad 2xy + \frac{y}{x^2 + y^2} = 3 \right\};$$

$$C = \left\{ (x, y); \quad x^3 - 3xy^2 + 3y = 1 \right\}; D = \left\{ (x, y); \quad 3x^2y - 3x - y^3 = 0 \right\}, \text{ then}$$

(A) $(A \cap B) = (C \cap D)$

(B) $(A \cap B) \neq (C \cap D)$

(C) $\frac{n(A \cap B)}{n(C \cap D)} = 1$

(D) $\frac{n(A \cap B)}{n(C \cap D)} = 4$

Ans. A, C

Sol. Let $z = x + iy$

A and B are real and imaginary part of $z^2 = z^{-1} + 3i$

Similarly C and D are real and imaginary part of $z^3 - 3iz = 1$

$$\Rightarrow (x, y) \in A \cap B \Leftrightarrow (x, y) \in C \cap D \Rightarrow A \cap B = C \cap D$$

40. Let $x_0 > 0 - \{3\}$ and x_n be defined recursively by $x_n = \sqrt{6 + x_{n-1}}$ where $n \geq 1$, then $\lim_{n \rightarrow \infty} x_n$ is

(A) $\sqrt{6}$

(B) $\sqrt{6 + x_0}$

(C) does not exist

(D) none of these

Ans. D

Sol. $x_n = \sqrt{6 + x_{n-1}}; \quad x_n^2 - x_{n-1}^2 = -(x_{n-1} - 3)(x_{n-1} + 2)$

Case-I :

$x_0 < 3$, then $x_n < 3$

Bounded by above

$$\text{Also, } x_n^2 - x_{n-1}^2 > 0$$

$x_n > x_{n-1}$ increasing sequence

Case-II :

$x_0 > 3$, then $x_2 > 3$

Bounded below

$$\text{So, } x_n^2 - x_{n-1}^2 < 0$$

$\Rightarrow x_n < x_{n-1}$ decreasing sequence

41. If ABC be a triangle inscribed in an ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$, then
- (A) area of $\triangle ABC$ is maximum when the eccentric angles of the vertices differ by $\frac{2\pi}{3}$
- (B) area of $\triangle ABC$ is maximum when the eccentric angles of the vertices differ by $\frac{5\pi}{9}$
- (C) maximum area of $\triangle ABC = 3\sqrt{\frac{3}{2}}$
- (D) if area of $\triangle ABC$ is maximum, then normals at A, B and C are concurrent

Ans. A, C

Sol. $\ar(\triangle ABC) = \frac{\ar(\triangle A'B'C')}{\sqrt{2}}$

Hence, two areas are maximum simultaneously thus $A'B'C$ is equilateral

So, $|\alpha - \beta| = |\beta - \gamma| = |\gamma - \alpha| = \frac{2\pi}{3}$; $\Delta(ABC) = 3\sqrt{\frac{3}{2}}$

Hence, let $\alpha < \beta < \gamma$; $\sin(\alpha + \beta) + \sin(\beta + \gamma) + \sin(\gamma + \alpha)$

$= \sin\left(\frac{2\pi}{3} + 2\alpha\right) + \sin(2\pi + 2\alpha) + \sin\left(\frac{4\pi}{3} + 2\alpha\right)$

$= 2\sin(\pi + 2\alpha)\cos\frac{\pi}{3} + \sin 2\alpha = 0$

42. If ABCD is a regular tetrahedron with length of any edge be 7, then

(A) volume of tetrahedron is $\frac{343}{6\sqrt{2}}$

(B) volume to tetrahedron is $\frac{243}{\sqrt{2}}$

(C) minimum distance of any vertex from the opposite face is $7\sqrt{\frac{2}{3}}$

(D) minimum distance of any vertex from the opposite face is $7\sqrt{\frac{3}{2}}$

Ans. A, C

Sol. $[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$; $\vec{a} \cdot \vec{a} = 7^2 = \vec{b} \cdot \vec{b} = \vec{c} \cdot \vec{c}$; $\vec{a} \cdot \vec{b} = 7^2 \times \cos 60^\circ$

$$v = \frac{1}{3}(\text{base area}) \times \text{height} ; h = \sqrt{\frac{2}{3}} \times 7$$

43. If $\sum_{n=0}^{\infty} \frac{(-1)^n}{3n+1} = \frac{1}{a} \left(\ln b + \frac{c}{\sqrt{a}} \right)$, then

(A) $a + b$ divides $(222)^{888} + (888)^{222}$

(B) $a + b$ is a prime number

(C) c is an irrational number

(D) $[2c] = 6$

(where $[.]$ denotes greatest integer function)

Ans. A, B, C, D

Sol. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$ (1)

$\omega^2 \ln(1+\omega x) = x - \frac{\omega x^2}{2} + \frac{\omega^2 x^3}{3} - \dots$ (2)

$\omega \ln(1+\omega^2 x) = x - \frac{\omega^2 x^2}{2} + \frac{\omega x^3}{3} - \frac{x^4}{4} + \dots$ (3)

Adding equation (1) + (2) + (3), we get $S = \frac{1}{3} \left(\ln 2 + \frac{\pi}{\sqrt{3}} \right)$

44. Which of the following are INCORRECT?

(A) $1999^{1999} < (2000)^{1998}$

(B) $e^{1/e} + e^{1/\pi} \leq 2e^{1/3}$

(C) $\log_2 3 < \log_3 5$

(D) $\frac{1}{\sqrt{4n}} < \left(\frac{1}{2}\right)\left(\frac{3}{4}\right)\dots\left(\frac{2n-1}{2n}\right) < \frac{1}{\sqrt{2n}}$

Ans. A, B, C

Sol. (A) Let $f(x) = (1999 - x) \ln(1999 + x)$

$f'(x) < 0 \forall 0 \leq x \leq 1$

$\Rightarrow f(0) > f(1)$

(B) $f(x) = e^{1/x}$, $f' < 0$, $f'' > 0$

f is dec and convex

(C) $x = \log_2 3$, $y = \log_3 5$

$2^x = 3$, $3^y = 5$

$27 = 3^3 = (2^x)^3 = 8^x$ and $25 = 5^2 = (3^y)^2 = 9^y$

$\Rightarrow 8^x > 9^y$ but $8 < 9 \Rightarrow x > y$

$$\begin{aligned}
 \text{(D) Let } P &= \left(\frac{1}{2}\right)\left(\frac{3}{4}\right)\dots\left(\frac{2n-1}{2n}\right) \\
 Q &= \frac{2}{3} \cdot \frac{4}{5} \dots \frac{2n-2}{2n-1} ; PQ = \frac{1}{2n} \\
 \therefore \frac{1}{2} &< \frac{2}{3} < \frac{3}{4} < \frac{4}{5} < \dots < \frac{2n-1}{2n} \\
 \text{So, } 2P &\geq Q ; 2P^2 \geq PQ \text{ (where } n \in \mathbb{I}^+)
 \end{aligned}$$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

If $f(x)$ be a continuous function over $[0, 1] \rightarrow \mathbb{R}^+$ such that $\int_0^1 f^2(x) dx = 1$ and $L = \int_0^1 x^{2021} f(x) dx$, then

45. Maximum value of L is

- (A) $\frac{1}{\sqrt{4041}}$
- (B) $\frac{2}{\sqrt{4041}}$
- (C) $\frac{1}{\sqrt{4043}}$
- (D) $\frac{3}{\sqrt{4043}}$

Ans. C

46. If maximum value of L is achieved for $f(x) = g(x)$, then number of points of non-differentiability of

$$\left(\left(\frac{x-1}{2} \right) \left(\frac{x-2}{3} \right) g(x) \right)^{\frac{1}{2023}} \quad \forall x \in (0, 1) \text{ is/are}$$

- (A) 0
- (B) 1
- (C) 2
- (D) none of these

Ans. C

Sol. (for Q. 45-46)

$$\left(\lim_{n \rightarrow \infty} \sum_{i=1}^n x_i^{2021} f(x_i) \Delta x \right)^2 \leq \left(\lim_{n \rightarrow \infty} \sum_{i=1}^n f^2(x_i) \Delta x \right) \left(\lim_{n \rightarrow \infty} \sum_{i=1}^n x_i^{4042} \Delta x \right)$$

$$= \int_0^1 f^2(x) dx \int_0^1 x^{4042} dx = \frac{1}{4043}$$

Paragraph for Question Nos. 47 and 48

Let A and B be points on the same branch of the hyperbola $xy = 1$. Suppose that P is a point lying between A and B on this hyperbola, such that the area of the triangle APB is maximum

47. Let $A \equiv (1, 1)$ and $B \equiv \left(4, \frac{1}{4}\right)$, then abscissa of P is equal to

- (A) $\sqrt{2}$
- (B) $\sqrt{3}$
- (C) 2
- (D) 3

Ans. C

48. $\frac{\text{Area of the region bounded by the hyperbola and chord AP}}{\text{Area of the region bounded by the hyperbola and chord PB}}$ is equal to

- (A) $\frac{1}{2}$
- (B) 1
- (C) 2
- (D) $\sqrt{3}$

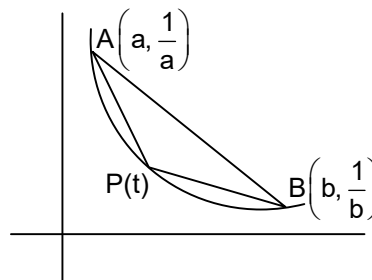
Ans. B

Sol. $A(t) = \frac{(a-b)(a-t)(b-t)}{2abt}$

$$A'(t) = \frac{(a-b)(t^2 - ab)}{2abt^2}$$

So, $t = \sqrt{ab}$ is point of maxima moreover

$$\int_a^{\sqrt{ab}} \left(-\frac{1}{at}x + \frac{1}{a} + \frac{1}{t} - \frac{1}{x} \right) dx = \int_{\sqrt{ab}}^b \left(-\frac{1}{bt}x + \frac{1}{b} + \frac{1}{t} - \frac{1}{x} \right) dx$$



Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

49. $\lim_{n \rightarrow \infty} \sin\left(\frac{1}{n^2}\right) + \sin\left(\frac{2}{n^2}\right) + \dots + \sin\left(\frac{n}{n^2}\right)$ is t , then $2t$ is equal to

Ans. 1

Sol. Let $\sin\left(\frac{1}{n^2}\right) + \sin\left(\frac{2}{n^2}\right) + \dots + \sin\left(\frac{n}{n^2}\right)$ be S_n

So, $\sum \frac{1}{n^2} - \frac{1}{6} \sum \frac{n^3}{n^6} < S_n < \frac{1}{n^2} + \frac{2}{n^2} + \dots + \frac{n}{n^2}$

50. If A is a 2×2 matrix such that $\text{trace}(A) = \det(A) = 3$, then $\text{trace}(A^{-1})$ is

Ans. 1

Sol. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $a + d = 3 = ad - bc$

51. To each positive integer with n^2 decimal digits we associate the determinant of the matrix obtained by writing the digits in order across the rows. For example, for $n = 2$, to the integer 8617 we associate $\det \begin{pmatrix} 8 & 6 \\ 1 & 7 \end{pmatrix} = 50$. If the sum of all the determinants associated with n^2 -digit integers be S , then

$\frac{S}{451 \times 45}$ is equal to

(Leading digits are assumed to be nonzero; for example, for $n = 2$, there are 9000 determinants)

Ans. 1

Sol. $n = 1 ; 1 + 2 + 3 + \dots + 9 = 45$

$n = 2 ; \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \dots + \begin{vmatrix} 9 & 9 \\ 9 & 9 \end{vmatrix}$

$= \sum_{\substack{1 \leq i \leq 9 \\ 0 \leq j, k, \ell \leq 9}} \begin{vmatrix} i & j \\ k & \ell \end{vmatrix} = \sum (i\ell - jk) = 100 \sum_{\substack{1 \leq i \leq 9 \\ 0 \leq \ell \leq 9}} i\ell - 90 \sum_{1 \leq j, k \leq 9} jk$

$= 100.45^2 - 90.45^2$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

Let ℓ be the equation of the line which intersect each of the two lines $x + 2y - 2 = 0 = 2x - 2y + z$ and $3x - 2y + z - 1 = 0 = x + 4y + z - 3$ and is parallel to the line $x + 2y - 3z = 0 = 2x + y + z + 3$. If line ℓ meets the plane $x + y + z = 0$ at the point (α, β, γ) , then

52. $|\alpha + \beta|$ is equal to

Ans. 0.80

53. $\left[\frac{\beta}{\gamma} \right]$ is equal to (where $[.]$ denotes greatest integer function)

Ans. 1.00

Sol. (for Q. 52-53)

Equation of a line which is coplanar to the lines $x + 2y - 2 + \lambda(2x - 2y + z) = 0$

$$(1 + 2\lambda)x + (2 - 2\lambda)y + \lambda z - 2 = 0 \quad \dots (1)$$

$$(3x - 2y + z - 1) + \mu(x + 4y + z - 3) = 0$$

$$(3 + \mu)x + (-2 + 4\mu)y + (1 + \mu)z + (-1 - 3\mu) = 0 \quad \dots (2)$$

$$(1) \text{ dot pd with } (5, -7, -3)$$

$$5 + 10\lambda - 14 + 14\lambda - 3\lambda = 0$$

$$21\lambda = 9 \therefore \lambda = \frac{3}{7}$$

$$(1) \text{ becomes, } \left(1 + \frac{6}{7}\right)x + \left(2 - \frac{6}{7}\right)y + \frac{3}{7}z - 2 = 0$$

$$\frac{13}{7}x + \frac{8}{7}y + \frac{3}{7}z - 2 = 0 ; 13x + 8y + 3z - 14 = 0 \quad \dots (3)$$

$$y = 1 ; 4x + 2y + 2z - 4 = 0$$

$$2x + y + z - 2 = 0 \quad \dots (4)$$

$$\text{So, line } \ell \text{ is } \frac{x - \frac{2}{5}}{\frac{2}{5}} = \frac{y - \frac{2}{3}}{-7} = \frac{z}{-3}$$

$$\text{Hence, } 5t + \frac{2}{3} - 7t + \frac{2}{3} - 3t = 0 \Rightarrow -5t = -\frac{4}{3} \Rightarrow t = \frac{4}{15}$$

$$\text{Point } \left(2, -\frac{6}{5}, -\frac{4}{5}\right)$$

Question Stem for Question Nos. 54 and 55
Question Stem

Suppose 20 distinct objects are placed along a circle at equal distances. Let n_1 be the number of ways of selecting 3 objects so that no two of them are adjacent and n_2 be the number of ways of selecting 3 objects so that neither of them are adjacent nor diametrically opposite, then

54. The value of $\frac{n_1}{400}$ is

Ans. 2.00

55. The value of $\frac{n_2}{75}$ is

Ans. 8.80

Sol. (for Q. 54-55)

$$n_1 = {}^{20}C_3 - 20 - 20 \times 16$$

$$n_2 = n_1 - ({}^{10}C_1)(14)$$

Question Stem for Question Nos. 56 and 57
Question Stem

The system of equations $2x - 3y + 5z = 12$, $3x + y + \lambda z = \mu$, $x - 7y + 8z = 17$ has

56. Will have a unique solution if $\lambda \neq k$, then k is equal to

Ans. 2.00

57. Will have infinite number of solutions if $\lambda + \mu$ is equal to

Ans. 9.00

Sol. (for Q. 56-57)

$$\Delta = \begin{vmatrix} 2 & -3 & 5 \\ 3 & 1 & \lambda \\ 1 & -7 & 8 \end{vmatrix} \neq 0, c_1 \rightarrow c_1 - c_2, \text{ then new } \Delta \text{ can be 0 if } \lambda = 2$$

Now if $\lambda = 2$

Then $2x - 3y + 5z = 12$, $3x + y + 2z = \mu$, $x - 7y + 8z = 17$ will have infinite solution if $\mu = 7$